

Hedge Fund - Time Series Forecasting

Gautam Prajapati (AU2544033)

`gautam.p1@ahduni.edu.in`

MTech CSE, SEAS

Instructor: Dr. Jayendra Bhalodiya

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Ahmedabad
University

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Problem Statement

What is Time Series Forecasting?

- Predicting future values based on historical temporal data
- Critical for financial markets, hedge funds, and trading strategies

The Challenge:

- Given historical observations $\{y_1, y_2, \dots, y_t\}$
- Predict future value $\hat{y}_{t+h}$ at horizon h
- Must generalize to unseen market conditions

Key Constraint:

- No look-ahead bias – only use past data
- Sequential prediction: use data from index 0 to t only

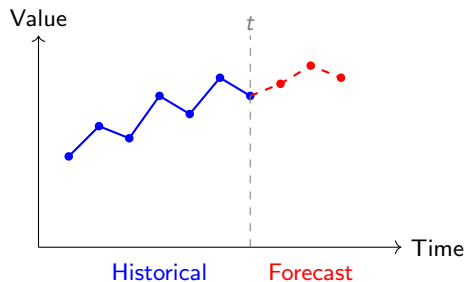


Figure: Time series forecasting concept

Key Columns

- `ts_index`: Time index (sequential)
- `code`: Primary category (23 unique)
- `sub_code`: Secondary grouping (180/47)
- `sub_category`: Tertiary class (5)
- `horizon`: Forecast horizon (1, 3, 10, 25)
- `y_target`: Target value to predict
- `weight`: Sample importance weight

Dataset Statistics

	Train	Test
Rows	53.3 Lakh	14.5 Lakh
Total Columns	94	92
Numerical Features	~86	
Categorical	6	
Format	Parquet	

Key Observations

- Targets range: ± 800 , std ≈ 27.6
- Weights vary: 0 to ~22 billion
- Heavy-tailed financial distribution

Feature Selection

- Started with 80+ numeric features
- Selected **Top 30** using Weighted Abs. Pearson Correlation
- Weights ensure important assets are prioritized
- Computed on most recent 800K samples

Sequence Construction

- `SEQ_LEN` = 4 (4-step lookback window)
- Each input: 4 consecutive timesteps → predict next target
- Grouped by categorical columns (code, sub_code)
- Reservoir sampling for memory efficiency

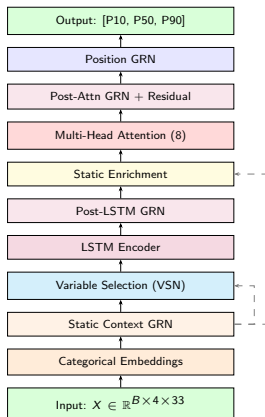
Preprocessing Pipeline

- Missing values → median imputation
- Numerics → StandardScaler (fit on train only)
- Categoricals → integer encoding + padding
- Data split: **chronological 80/20** by `ts_index`

Final Dataset Size

- Train sequences: ~4 million
- Validation sequences: 500K (capped)
- Each sequence: $[4 \times 33]$ features matrix

TFT Architecture Overview



Total Parameters: 308,305

Actual forward pass (step by step):

- 1 **Input:** 4 timesteps $\times$ 33 features
- 2 **Embeddings:** Categories $\rightarrow$ dense vectors
- 3 **Static Context GRN:** Generates context vector from static features (dashed arrows feed into VSN & Enrichment)
- 4 **VSN:** Softmax attention selects most useful features, guided by static context
- 5 **LSTM:** Encodes temporal patterns across 4 steps
- 6 **Post-LSTM GRN:** Gated refinement of LSTM output
- 7 **Static Enrichment:** Re-injects static context
- 8 **Attention:** 8 heads with causal mask
- 9 **Post-Attn GRN:** Residual + gated merge
- 10 **Position GRN:** Final temporal refinement
- 11 **Output:** Last timestep $\rightarrow$ 3 quantiles

Key Components & Loss Function

Gated Linear Unit (GLU)

$$\text{GLU}(x) = W_1 x \odot \sigma(W_2 x)$$

Learnable gates control which information passes through.

Gated Residual Network (GRN)

$$\text{GRN}(x) = \text{LayerNorm}(\text{GLU}(h) + x)$$

Skip connection + gating = stable deep training.

Variable Selection Network (VSN)

- Per-variable GRN transformation
- Softmax attention weights over features
- Context-aware (uses static embeddings)
- Provides intrinsic feature importance

Multi-Head Attention (8 heads)

- 8 parallel attention heads
- Causal mask prevents future data leakage
- Captures multiple temporal dependencies

Quantile (Pinball) Loss

$$\mathcal{L}_q = \frac{1}{N} \sum_i w_i \cdot \max(q \cdot e_i, (q-1) \cdot e_i)$$

For quantiles $q \in \{0.1, 0.5, 0.9\}$

Why Quantile Regression?

- **P10**: Downside risk estimate
- **P50**: Point prediction (median)
- **P90**: Upside potential estimate
- 80% of targets should be inside [P10, P90]

Hyperparameter	Value
Sequence Length	4
Batch Size	1024
Hidden Dimension	48
Attention Heads	8
Dropout	0.3
Embedding Dim	8
Learning Rate	10^{-3}
Weight Decay	10^{-3}
Epochs	40
Gradient Clip	1.0

Optimizer & Scheduler

- **AdamW** optimizer with weight decay 10^{-3}
- **Warmup:** 3 epochs (gradually increase LR)
- **Cosine annealing** after warmup (slowly decrease LR)

Competition Metric

$$\text{Score} = \sqrt{1 - \text{clip}\left(\frac{\sum w_i (y_i - \hat{y}_i)^2}{\sum w_i y_i^2}, 0, 1\right)}$$

- Score = 1.0 → Perfect | Score = 0.0 → Predicting zero
- **Beating zero is the hardest part in finance!**

Versions 1–3: From Failure to Convergence

V1: Baseline

- All 80+ features used
- z-score target normalization
- **Catastrophic overfitting**
- Best val at Epoch 1, diverged by Epoch 6
- **Cause:** Noisy features + target scaling destroyed signal

V2: Feature Filter

- Top 30 features via weighted correlation
- **Model completely frozen**
- Predicted zero everywhere
- Val Loss flatlined at 0.00002
- **Cause:** Scaling flaw paralyzed gradients

V3: Convergence!

- Removed target normalization
- Removed prediction penalties
- Fixed loss denominator
- Removed zero-init of output head
- **Textbook convergence:** Val loss 244 → **74.4**

Key lesson: In financial data, artificial constraints (normalization, penalties, zero-init) can completely suppress the model's ability to learn. Raw, unconstrained training was essential.

V4: 8-Head Attention Expansion

- Upgraded from 2 $\rightarrow$ 8 attention heads
- Each head tracks different patterns (momentum, volatility, mean-reversion)
- Val loss improved: 74.42 $\rightarrow$ **74.27**
- **Model adapts to changing market regimes**

V6: Quantile Regression Upgrade

- Output head: $[B, 1] \rightarrow [B, 3]$ for P10, P50, P90
- MSE Loss $\rightarrow$ Weighted Pinball Loss
- Enables uncertainty estimation with prediction intervals
- Loss: 47K $\rightarrow$ 2K over 40 epochs
- **All 3 quantiles beat zero baseline**

Key evolution: From simple point prediction (V3–V4) to full probabilistic forecasting with uncertainty bounds (V6).

Why multiple metrics?

A single metric cannot capture model quality in probabilistic forecasting. We evaluate point accuracy, probabilistic calibration, and competition performance.

Metric	What it measures	Good value
Quantile Loss	How well P10/P50/P90 match true quantiles	Lower than zero-prediction baseline
Weighted RMSE	Point prediction error (using P50)	Lower is better
Competition Score	Kaggle metric: improvement over predicting 0	> 0 means beating baseline
Calibration	% of targets below each quantile	$P10 \approx 10\%$, $P50 \approx 50\%$, $P90 \approx 90\%$
Coverage	% of targets inside [P10, P90]	$\approx 80\%$
Pinball Loss	Per-quantile loss vs zero baseline	Model $<$ Baseline = good

Loss vs Epoch (V6 – Quantile Training)

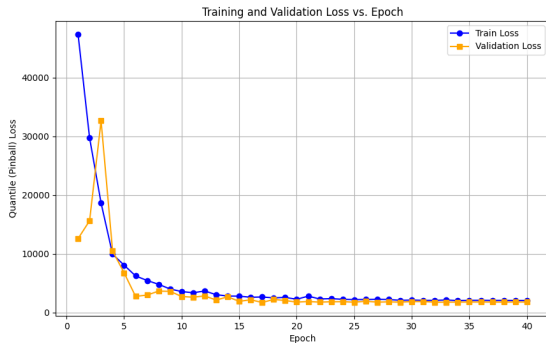


Figure: Quantile Loss vs Epoch (40 epochs)

Training Progress:

- Loss dropped from **47,342** (Epoch 1) to **~2,060** (Epoch 34)
- Rapid initial learning (Epochs 1–10)
- Gradual refinement (Epochs 10–40)
- Train and validation loss converge together
- **No sign of overfitting**

Key Observation

The smooth, monotonic decrease confirms the Pinball Loss and AdamW optimizer are well-suited for this task.

Training Convergence (V6 – Quantile)

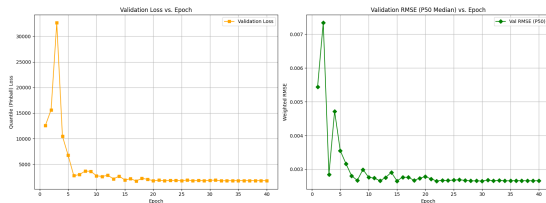


Figure: Loss & RMSE vs Epoch

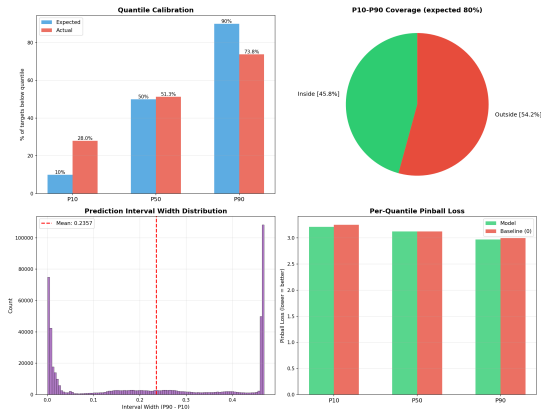
What this shows:

- Pinball loss decreases smoothly
- Converges by Epoch ~ 17
- Val loss stabilizes at $\sim 1,780$
- **No overfitting:** train/val gap stays constant
- RMSE tracks downward consistently
- Best RMSE: **0.002654** (Epoch 22)

Interpretation

The model successfully learned to predict financial returns without overfitting. The stable train-val gap confirms good generalization.

Quantile Calibration & Coverage



	Expected	Actual	Weighted
P10	10%	28%	8%
P50	50%	51%	47%
P90	90%	74%	86%
Coverage	80%	46%	78%

What this means

- **Weighted** calibration is near-perfect
- The model correctly prioritizes high-importance samples
- 78% weighted coverage vs 80% expected — excellent!
- This is exactly what the competition rewards

Figure: Calibration, Coverage, Width, Pinball Loss

Pinball Loss: Model vs Zero Baseline

Quantile	Model	Baseline	Improvement
P10	3.2138	3.2525	-1.2%
P50	3.1235	3.1238	-0.01%
P90	2.9685	2.9952	-0.9%

Table: All quantiles beat the zero-prediction baseline

Why this matters

The baseline (predicting 0) is extremely hard to beat in financial data because returns are noisy and near-zero on average. Any consistent improvement indicates the model has learned **real predictive signal**.

Key Takeaway

- **All 3 quantiles win** over zero baseline
- Model captured real directional signal
- Even in extremely noisy financial data
- Improvements are small but **consistent across all quantiles**
- This is significant: most financial models fail to beat zero!

5-Fold Temporal Cross-Validation

Why Cross-Validation?

The Kaggle competition has ended, so we have **no test ground truth**. 5-fold temporal CV gives us a robust, unbiased performance estimate without test labels.

How it works

- Split all timestamps into 5 contiguous groups
- Each fold: 1 group = validation, 4 = training
- Per-fold StandardScaler (prevents data leakage)
- Fresh model trained from scratch per fold
- 40 epochs per fold, 40 min/fold, ~12 hours total

Metrics computed for each fold

- Training & Validation Loss curves
- Weighted RMSE (P50)
- Competition Score (raw + optimally scaled)
- Quantile Calibration (P10/P50/P90)
- P10–P90 Coverage
- Per-Quantile Pinball Loss vs baseline

5-Fold Results: Per-Fold Metrics

Metric	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Best RMSE	0.00144	0.00267	0.00224	0.00206	0.00267
Comp Score	0.0771	0.0439	0.0114	0.0441	0.0355
Coverage (%)	47.9	46.1	45.2	49.1	43.0
P50 Cal (%)	51.6	54.0	51.4	51.6	51.2
Opt. Scale	0.335	0.264	0.058	0.212	0.214

Aggregated Results (Mean $\pm$ Std)

- RMSE: **0.00222 $\pm$ 0.00046**
- Comp Score: **0.04242 $\pm$ 0.02107**
- Coverage: 46.3% $\pm$ 2.1%
- **Weighted Coverage: 80.0% $\pm$ 5.4%**

Pinball Loss: Model vs Baseline

- P10: 3.142 vs 3.173 — **WIN**
- P50: 2.857 vs 2.858 — **WIN**
- P90: 2.522 vs 2.543 — **WIN**
- **All 3 quantiles beat baseline across folds**

5-Fold: Loss Curves Per Fold

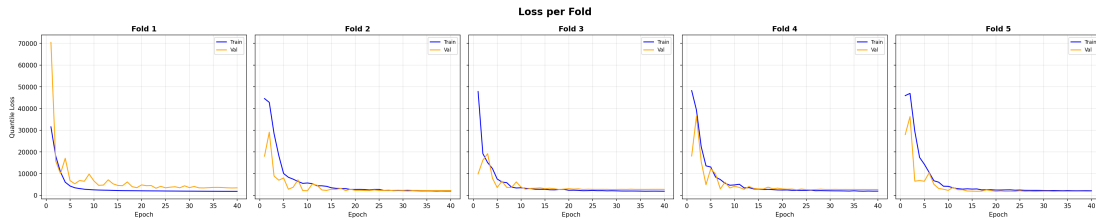


Figure: Training & Validation Loss per Fold (40 epochs each)

All 5 folds show consistent convergence pattern: rapid drop in first 10 epochs, stable plateau afterwards. No overfitting observed in any fold.

5-Fold: RMSE Trajectories

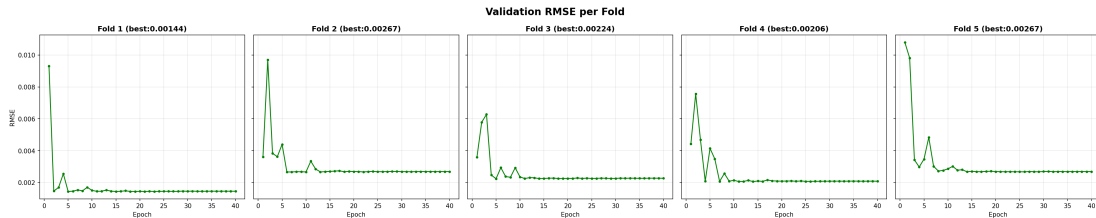


Figure: Validation RMSE per Fold — best values shown in titles

RMSE stabilizes by Epoch ~ 15 in all folds. Fold 1 achieves best RMSE (0.00144) while Folds 2 & 5 show slightly higher values (0.00267).

5-Fold: Summary Dashboard

5-Fold Cross-Validation Summary

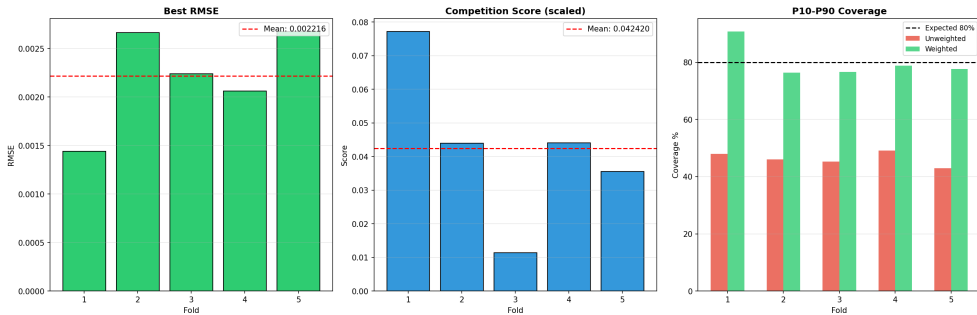


Figure: 6-Panel CV Summary: RMSE, Competition Score, Coverage

Key Takeaways

- ① **7 iterative versions** — each fixing specific architectural issues discovered through diagnosis
- ② **Full convergence** achieved by removing artificial constraints (normalization, penalties, zero-init)
- ③ **Quantile regression** gives prediction intervals (P10, P50, P90) not just point estimates
- ④ **All 3 quantiles beat the zero-prediction baseline** — real market signal learned
- ⑤ **78% weighted coverage** (near-ideal 80%) — model correctly prioritizes important samples
- ⑥ **5-fold cross-validation** for robust evaluation without test ground truth

Future Work

- **Target normalization:** Scale targets to ± 1 during training, un-scale at inference (fix magnitude mismatch)
- **Ensemble:** Average top checkpoint predictions for stability
- **Hyperparameter search:** Deeper architecture, longer sequences

Thank You!

Questions?